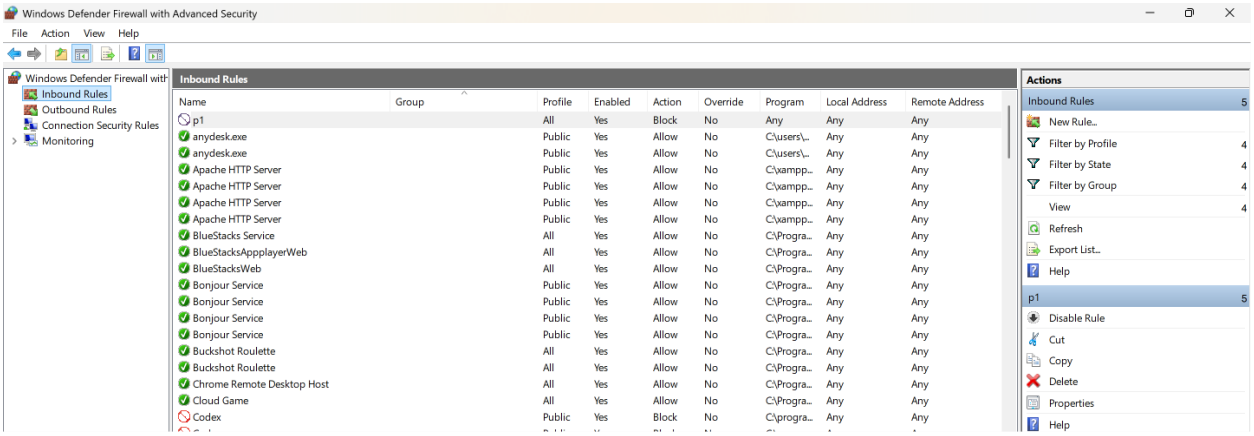
Secure networks with firewalls NAT, port filtering, packet filtering, and other methods

1.Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

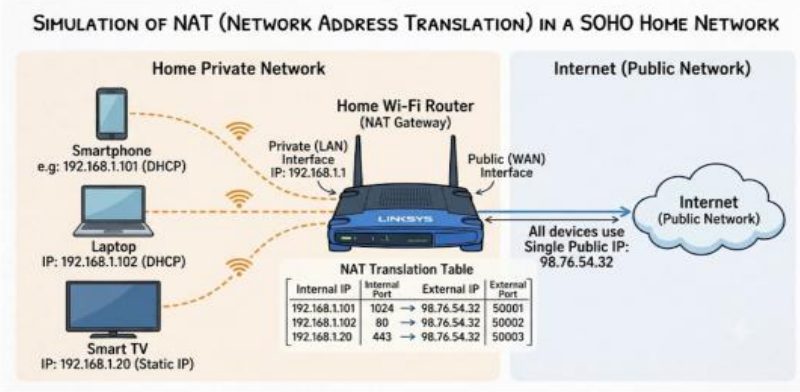
Steps to Block Port 8080

1. Press Windows + R, type wf.msc, and press Enter.
2. Select Inbound Rules → New Rule.
3. Choose Port → TCP and enter 8080.
4. Select Block the connection.
5. Select Domain, Private, and Public.
6. Name the rule Block Incoming Port 8080 and click Finish.



**2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.**



3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Steps to Block Outgoing Port 25

1. Open your router’s Admin Panel (usually 192.168.1.1).
2. Log in using the router admin credentials.
3. Go to Firewall / Security → Port Filtering.
4. Select Add New Rule.
5. Set Protocol: TCP, Port: 25, and Direction: Outgoing.
6. Select Block/Deny and save the settings.

**4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.**

1. HTTP: 192.168.1.5 → 172.217.166.78:80 — Allowed - Port 80 is used for web traffic and is generally permitted.

2. SSH: 10.0.0.10 → 34.201.23.1:22 — Blocked — Port 22 is used for SSH and is commonly blocked to prevent unauthorized remote access

3. DNS: 192.168.1.7 → 8.8.8.8:53 — Allowed — Port 53 is used for DNS and is required for domain name resolution.

5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

1. Block malicious IP addresses — Firewalls can block traffic from known harmful or suspicious IP addresses to reduce attacks.

2. Allow only required ports — Packet filtering can allow necessary ports such as HTTPS (443) while blocking unnecessary ports.

3. Prevent suspicious traffic — Firewalls can filter unusual or excessive traffic to help protect servers from DDoS attacks and unauthorized access.